

Highway Substrate Audit Summary

Prepared by Cyfrin

Version 2.1

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Highway on Highway Substrate. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.1 dated September 29, 2026, was delivered to Highway, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 15 days, the Cyfrin team conducted an audit on the Highway Substrate code provided by Highway. The findings consist of 1 Critical, 5 Medium & 32 Low severity issues with the remainder being informational.

Summary

Project Name	Highway Substrate
Repository	hway-substrate
Commit	df99567ea908...
Audit Timeline	June 29th - July 17th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Partially Resolved	Acknowledged
Critical Risk	1	1	0	0
High Risk	0	0	0	0
Medium Risk	5	3	0	2
Low Risk	32	20	1	11
Informational	1	1	0	0
Gas Optimizations	0	0	0	0
Total	39	25	1	13

6 Audit Scope

The audit scope was limited to:

```
pallets/highway-bls-utils/src/lib.rs
pallets/highway-config/src/lib.rs
pallets/highway-entry/src/lib.rs
pallets/highway-nft-delegation-registry/src/lib.rs
pallets/highway-nft-permission/src/lib.rs
pallets/highway-nft-types/src/lib.rs
pallets/highway-registry/src/lib.rs
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Highway (the Mosaic Bridge) is a cross-chain token-and-message bridge. This summary covers its Substrate implementation; the bridge also has EVM (Ethereum) and Solana implementations that act as this runtime's cross-chain counterparties. A message emitted on a source chain is attested by a BLS12-381 committee drawn from a shared relayer set and executed on the destination chain, where it mints or releases tokens and/or dispatches a payload call.

8.1 Components

Pallet	Role
`highway-entry`	Inbound and outbound entrypoints: `emit_message` (burn/escrow, `messageId` assignment) and `receive_message` (BLS aggregate verification, committee selection, mint/release, payload dispatch via `execute_payload`)
`highway-registry`	Relayer registration, BLS keys, operational keys, registered-relayer bitmap, active-set epochs
`highway-config`	Token and corridor configuration, decimal handling, payload whitelist
`highway-nft-delegation-registry`	Relayer backing and delegation: operator self-binding, external delegation, tier ceilings, slashing, undelegation
`highway-nft-types` / `highway-nft-permission`	NFT tier definitions (Nomad / Anchor / Vanguard capacity tiers) and permission-token minting
`highway-bls-utils`	BLS12-381 primitives, EIP-2537 byte-format helpers, hash-to-curve support

8.2 Message Lifecycle

1. **Outbound (source).** The user calls `emit_message` with a target chain, an optional token leg (token, target address, amount) and an optional payload leg (target address, encoded call), plus fee handling. The source debits the funds (burn or escrow), allocates a nonce, computes the `messageId`, and emits the message.
2. **Attestation (off-chain committee).** The relayers seated in the active committee for the message's slot co-sign the message and produce a single BLS aggregate signature plus a committee bitmap.
3. **Inbound (destination).** A committee relayer submits the attestation via `receive_message`, which re-derives the `messageId`, verifies the aggregate signature against the committee it selects for that slot, records the message as processed, then performs the token leg (mint or release) and the payload leg (`execute_payload` dispatches the payload call).

Each (token, chain) corridor is configured as either burn and mint or escrow and release. Committee public keys are in G1 and signatures in G2; the runtime uses the platform's BLS12-381 primitives.

9 Attestation

Cyfrin attests that it performed a security review of the Highway Substrate code provided by Highway.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-09-29-cyfrin-highway-substrate-v2.1.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Highway has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=e0de8b06cf07687ab7196904f4bf2f543e9a388e31c13aaf9113a24492cffc90
F=2026-09-29-cyfrin-highway-substrate-v2.1-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.